



(Liquid + liquid) equilibria for the ternary mixtures (alkane + toluene + ionic liquid) at $T = 298.15$ K: Influence of the anion on the phase equilibria

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ABSTRACT

(Liquid + liquid) equilibrium data for the ionic liquids 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide, [BMpyr][NTf₂], and 1-butyl-1-methylpyrrolidinium trifluoromethanesulfonate, [BMpyr][TFO], with toluene, and heptane or cyclohexane were determined at $T = 298.15$ K and atmospheric pressure. In order to check if these ILs can be used as potential solvents for the extraction of toluene from aliphatic compounds, the ability of the ILs as solvents was evaluated in terms of selectivity and solute distribution ratio. The experimental data were correlated accurately with the Non Random Two-Liquid model.

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1. Introduction

In the oil refining industry, extraction of aromatics from naphta, fuel jets, and kerosene, is one of the major importance processes. This separation cannot be made through distillation process, due to the proximity of the boiling points of the hydrocarbons and the presence of azeotropic mixtures, therefore liquid extraction is used in the industry to separate this kind of components. The most common processes use solvents such as sulfolane, glycols, dimethylsulfoxide, N-methylpyrrolidone [1], etc.

As the design of safe and environmentally friendly separation processes is increasing its importance in the chemical industry, over the last years ionic liquids (ILs) were suggested as an alternative to traditional organic solvents for extraction processes because they show interesting properties which would produce less damage to the environment and improving performance [2–5].

As a continuation of our ongoing work on aromatic extraction using ILs [6–10], we analyzed the suitability of the ILs [BMpyr][NTf₂] and [BMpyr][TFO] to separate aromatic compounds in the ternary mixtures {heptane, or cyclohexane (1) + toluene (2) + [BMpyr][NTf₂], or [BMpyr][TFO] (3)} and {heptane (1) + cyclohexane (2) + [BMpyr][NTf₂], or [BMpyr][TFO] (3)} at $T = 298.15$ K. Besides, scarce published works on (liquid + liquid) equilibria (LLE) comparing two

different anions in ILs with the same cation are published. The values obtained were compared with other published results concerning sulfolane and ILs. In addition, the NRTL model [11] was used to correlate the experimental LLE data for the six studied ternary systems.

2. Experimental

2.1. Chemicals

Cyclohexane, heptane and toluene were supplied by Sigma-Aldrich. They were degassed ultrasonically, dried over molecular sieves of $4 \cdot 10^{-8}$ cm Aldrich and kept in an inert argon atmosphere. The 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide, [BMpyr][NTf₂], and 1-butyl-1-methylpyrrolidinium trifluoromethanesulfonate, [BMpyr][TFO], were purchased from IoLiTec with a mass fraction of water of $6 \cdot 10^{-4}$ for [BMpyr][NTf₂] and $2.7 \cdot 10^{-3}$ for [BMpyr][TFO], and a mass fraction of halides lower than $1 \cdot 10^{-3}$. In order to reduce the water content to negligible values, since they have an important influence over the physical properties of the ILs [12], the ILs were dried with stirring at moderate temperature (343 K) and under vacuum (0.2 Pa) for at least 48 h prior to their use. The water content after the drying step was found to be in the same order that the initial data provided by IoLiTec. After the drying the ILs were kept in bottles in a glove box under an inert gas.

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The experimental physical properties (density, ρ , and refractive index, n_D) of the pure components at $T = 298.15$ K and atmospheric pressure, just as provenance and mass fraction purity, are presented in table 1, together with recent literature data [13–15].

2.2. Apparatus and procedure

LLE data were determined at $T = 298.15$ K and atmospheric pressure. The determination of the binodal curves, validation points, and tie-lines are explained in detail in a previous paper [8]. Briefly, solubility curves were obtained using the cloud point method [16], while tie-line compositions were determined by density measurements. For the preparation of samples, a Mettler AX-205 Delta Range balance with an uncertainty of $\pm 3 \cdot 10^{-4}$ g was used. The density of the samples was measured using an Anton Paar DSA-5000M digital vibrating tube densimeter with an uncertainty in the experimental measurements of $\pm 3 \cdot 10^{-5}$ g · cm⁻³.

The binodal curves of the systems studied were determined at $T = 298.15$ K and atmospheric pressure by titrating binary mixtures of known compositions with the third component until the transition was visually found. The density of each sample was measured and a polynomial expression for the density as function of composition was obtained. To estimate the error of the technique used for the determination of compositions, the density of three new points, prepared by weighing (validation points), was determined for each system, and with this density value, by means of the above-mentioned polynomial expression, the composition was

TABLE 1

Supplier, purity, density, ρ , and refractive index, n_D , of pure components at $T = 298.15$ K.

Component	Supplier	Mass fraction purity	ρ /(g · cm ⁻³)		n_D	
			Exp.	Lit.	Exp.	Lit.
[BMpyr][NTf ₂]	IoLiTec	>0.999	1.39466	1.39459 ^a	1.42302	1.42304 ^a
[BMpyr][TFO]	IoLiTec	>0.999	1.25271	n.a.	1.43287	n.a.
Heptane	Sigma–Aldrich	>0.999	0.67957	0.67946 ^b	1.38529	1.38511 ^b
Cyclohexane	Sigma–Aldrich	>0.995	0.77388	0.77386 ^b	1.42356	1.42354 ^b
Toluene	Sigma–Aldrich	>0.999	0.86223	0.86219 ^b	1.49398	1.49390 ^c

n.a. not available.

Standard uncertainty: ρ is ± 0.00003 g · cm⁻³ and n_D is ± 0.00004 .

^a Reference [13].

^b Reference [14].

^c Reference [15].

TABLE 2

Density data for binary system of {heptane (1) + cyclohexane (2)} at $T = 298.15$ K.

x_1	ρ /(g · cm ⁻³)
1.0000	0.67957
0.9539	0.68260
0.9061	0.68562
0.8061	0.69271
0.7062	0.70061
0.6660	0.70379
0.5039	0.71721
0.3926	0.72661
0.3118	0.73569
0.2021	0.74816
0.0991	0.76053
0.0503	0.76695
0.0000	0.77388

Standard uncertainty: x_1 is ± 0.0001 and ρ is ± 0.00003 g · cm⁻³

TABLE 3

Experimental (liquid + liquid) equilibrium data in mole fraction for the studied ternary systems at $T = 298.15$ K, solute distribution ratio (β), and selectivity (S) values.

Upper phase		Lower phase		β	S
x_1^I	x_2^I	x_2^{II}	x_1^{II}		
Heptane (1) + toluene (2) + [BMpyr][NTf ₂] (3)					
1.000	0.000	0.064	0.000		
0.965	0.035	0.065	0.035	0.98	14.52
0.906	0.094	0.062	0.101	1.07	15.69
0.864	0.136	0.061	0.149	1.10	15.71
0.811	0.189	0.059	0.194	1.03	14.13
0.755	0.245	0.057	0.238	0.97	12.94
0.709	0.291	0.056	0.273	0.94	11.82
0.654	0.346	0.057	0.306	0.88	10.20
0.601	0.399	0.052	0.352	0.88	10.12
0.479	0.521	0.051	0.421	0.81	7.56
0.351	0.649	0.043	0.490	0.76	6.12
0.232	0.768	0.042	0.556	0.72	3.98
0.000	1.000	0.000	0.734		
Cyclohexane (1) + toluene (2) + [BMpyr][NTf ₂] (3)					
1.000	0.000	0.294	0.000		
0.945	0.055	0.251	0.048	0.87	3.28
0.879	0.121	0.214	0.107	0.88	3.60
0.814	0.186	0.186	0.168	0.90	3.95
0.758	0.242	0.162	0.238	0.99	4.63
0.697	0.303	0.147	0.289	0.95	4.52
0.643	0.357	0.136	0.328	0.92	4.34
0.584	0.416	0.125	0.367	0.88	4.11
0.443	0.557	0.097	0.471	0.85	3.87
0.327	0.673	0.079	0.529	0.79	3.25
0.161	0.839	0.044	0.631	0.75	2.73
0.000	1.000	0.000	0.734		
Heptane (1) + cyclohexane (2) + [BMpyr][NTf ₂] (3)					
1.000	0.000	0.064	0.000		
0.920	0.080	0.058	0.018	0.23	3.58
0.847	0.153	0.051	0.034	0.22	3.66
0.794	0.206	0.049	0.046	0.22	3.59
0.710	0.290	0.044	0.068	0.23	3.73
0.624	0.376	0.038	0.088	0.23	3.81
0.516	0.484	0.034	0.111	0.23	3.45
0.398	0.602	0.031	0.134	0.22	2.87
0.000	1.000	0.000	0.294		
Heptane (1) + toluene (2) + [BMpyr][TFO] (3)					
1.000	0.000	0.032	0.000		
0.925	0.075	0.031	0.067	0.89	26.68
0.837	0.163	0.028	0.131	0.80	23.96
0.751	0.249	0.027	0.184	0.74	20.45
0.660	0.340	0.024	0.236	0.69	18.96
0.560	0.440	0.023	0.291	0.66	15.88
0.456	0.544	0.023	0.339	0.62	12.39
0.353	0.647	0.023	0.382	0.59	9.20
0.203	0.797	0.022	0.483	0.61	5.64
0.000	1.000	0.000	0.520		
Cyclohexane (1) + toluene (2) + [BMpyr][TFO] (3)					
1.000	0.000	0.075	0.000		
0.919	0.081	0.081	0.065	0.80	9.14
0.832	0.168	0.080	0.131	0.78	8.03
0.741	0.259	0.078	0.182	0.70	6.62
0.647	0.353	0.074	0.243	0.69	6.02
0.545	0.455	0.068	0.285	0.63	5.04
0.436	0.564	0.059	0.339	0.60	4.41
0.331	0.669	0.049	0.393	0.59	3.93
0.184	0.816	0.031	0.475	0.58	3.51
0.000	1.000	0.000	0.520		
Heptane (1) + cyclohexane (2) + [BMpyr][TFO] (3)					
1.000	0.000	0.032	0.000		
0.908	0.092	0.023	0.029	0.32	12.71
0.807	0.193	0.020	0.039	0.20	8.17
0.703	0.297	0.017	0.048	0.16	6.56
0.607	0.393	0.016	0.052	0.13	4.95
0.497	0.503	0.015	0.054	0.11	3.45
0.391	0.609	0.015	0.057	0.09	2.50
0.290	0.710	0.014	0.057	0.08	1.63
0.187	0.813	0.013	0.060	0.07	1.10
0.000	1.000	0.000	0.075		

determined. Comparing the experimental and calculated compositions, the maximum error was estimated to be 0.007 in mole fraction.

For the tie-line determination, ternary mixtures of known composition inside the immiscible region were placed into glass cells and kept in a thermostatic bath. The temperature was maintained

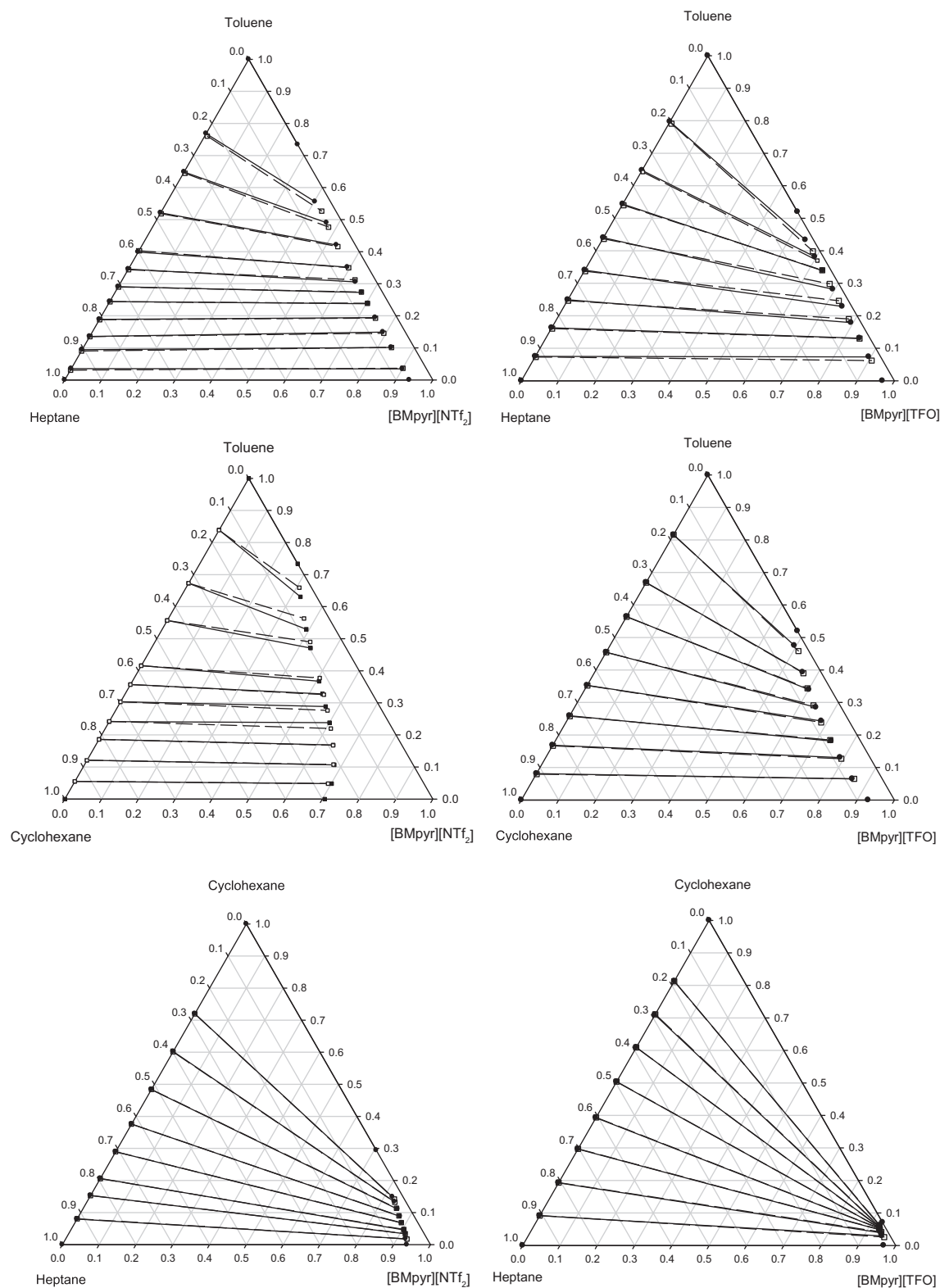


FIGURE 1. Experimental LLE for the ternary systems (a) {heptane (1) + toluene (2) + [BMpyr][NTf₂] (3)}; (b) {heptane (1) + toluene (2) + [BMpyr][TFO] (3)}; (c) {cyclohexane (1) + toluene (2) + [BMpyr][NTf₂] (3)}; (d) {cyclohexane (1) + toluene (2) + [BMpyr][TFO] (3)}; (e) {heptane (1) + cyclohexane (2) + [BMpyr][NTf₂] (3)}; (f) {heptane (1) + cyclohexane (2) + [BMpyr][TFO] (3)} at $T = 298.15$ K. Solid lines and full points represent experimental tie-lines, and dashed lines and empty squares indicate calculated data from NRTL model.

constant in a thermostatic bath PoliScience digital temperature controller with an uncertainty of ± 0.01 K. The mixtures were vigorously stirred for 5 h in order to allow an intimate contact between phases, and then settled overnight to guarantee a complete split of the equilibrium phases. A sample from each phase was taken with a syringe for composition analysis by means of densities. Some samples of the upper phase (alkane-rich phase) were analyzed by ^1H NMR, and since no ionic liquid was found, the compositions were obtained using the density versus composition data at $T = 298.15$ K of the binary systems {heptane (1) + toluene (2)} and {cyclohexane (1) + toluene (2)}, taken from literature [7,17], and {heptane (1) + cyclohexane (2)}, determined in this work and presented in table 2.

Since the composition of the third component can be written in terms of the other two, and taking into account that each point of the solubility curve shows a different density value, one property is enough to determine the composition of the lower-rich phase (IL-rich phase).

3. Results and discussion

LLE data for the ternary mixtures {heptane, or cyclohexane (1) + toluene (2) + [BMpyr][NTf₂], or [BMpyr][TFO] (3)} and {heptane (1) + cyclohexane (2) + [BMpyr][NTf₂], or [BMpyr][TFO] (3)} at $T = 298.15$ K are reported in table 3. The corresponding triangular diagrams of the studied systems are given in figure 1. As can be seen in this figure, the systems studied present a wide immiscibility region, one pair of components show total miscibility and two of their constituent pairs show partial immiscibility. Comparing the size of the miscible region, it is worth noting that the anion has an important influence on the solubility. Comparing ILs with the same [BMpyr]⁺ cation, the IL with the [TFO][−] anion as a bigger immiscible region than that with the [NTf₂][−] anion. As can be inferred from table 3, toluene presents partial solubility in the two tested ILs and in the order [BMpyr][NTf₂] > [BMpyr][TFO], with $x_2 = 0.734$, and $x_2 = 0.520$, respectively. The solubility of cyclohexane and heptane in the ILs is also in the order [BMpyr][NTf₂] > [BMpyr][TFO], but it is lower than for toluene. The maximum solubility for cyclohexane is $x_1 = 0.294$ in [BMpyr][NTf₂] and $x_1 = 0.075$ in [BMpyr][TFO], while for heptane $x_1 = 0.064$, and $x_1 = 0.032$, respectively.

3.1. Solute distribution and selectivity

The solvent ability of the ILs studied for the separation of toluene from heptane or cyclohexane was analyzed by terms of solute distribution ratio, β , and selectivity, S . Together with the LLE experimental data, table 3 includes the corresponding values, defined as:

$$\beta = \frac{x_2^{\text{II}}}{x_2^{\text{I}}}, \quad (1)$$

$$S = \frac{x_2^{\text{II}} x_1^{\text{I}}}{x_1^{\text{II}} x_2^{\text{I}}}, \quad (2)$$

where x_1^{I} and x_2^{I} are the mole fractions of alkane and toluene, in the upper phase (alkane-rich phase), respectively; and x_1^{II} and x_2^{II} are the mole fractions of alkane and toluene, in the lower phase (IL-rich phase), respectively. Notice that for the system {heptane (1) + cyclohexane (2) + IL (3)} x_1^{I} and x_2^{I} are the mole fractions of heptane and cyclohexane in the upper phase, respectively, and x_1^{II} and x_2^{II} the corresponding mole fractions in the lower phase.

In figure 2 are shown the S values for the ternary systems {heptane (1) + toluene (2) + solvent (3)} and {cyclohexane (1) + toluene (2) + solvent (3)} at $T = 298.15$ K as a function of toluene mole fraction in the alkane-rich phase. It can be observed that the determined values of S decrease when the content of toluene increases and that S values are higher than unity, showing that the studied ILs are suitable for use in solvent extraction. The systems with the IL [BMpyr][TFO] present the higher S values, similar to those obtained with sulfolane. The system {heptane (1) + toluene (2) + [BMpyr][NTf₂] (3)} was compared with those published by Pereiro and Rodríguez [13], showing a good agreement with the data reported by these authors. In figure 2, the S values calculated from the NRTL model are plotted. Comparing the data obtained with those calculated with the NRTL model we can conclude that, in general, the model correlates adequately the experimental values, since the greater deviations are found for the system {cyclohexane (1) + toluene (2) + [BMpyr][NTf₂] (3)} in the lower composition range.

In figure 3 the β values for the ternary systems {heptane (1) + toluene (2) + solvent (3)} and {cyclohexane (1) + toluene (2) + solvent (3)} at $T = 298.15$ K as a function of the toluene mole fraction and mass fraction in the alkane-rich phase are plotted. As

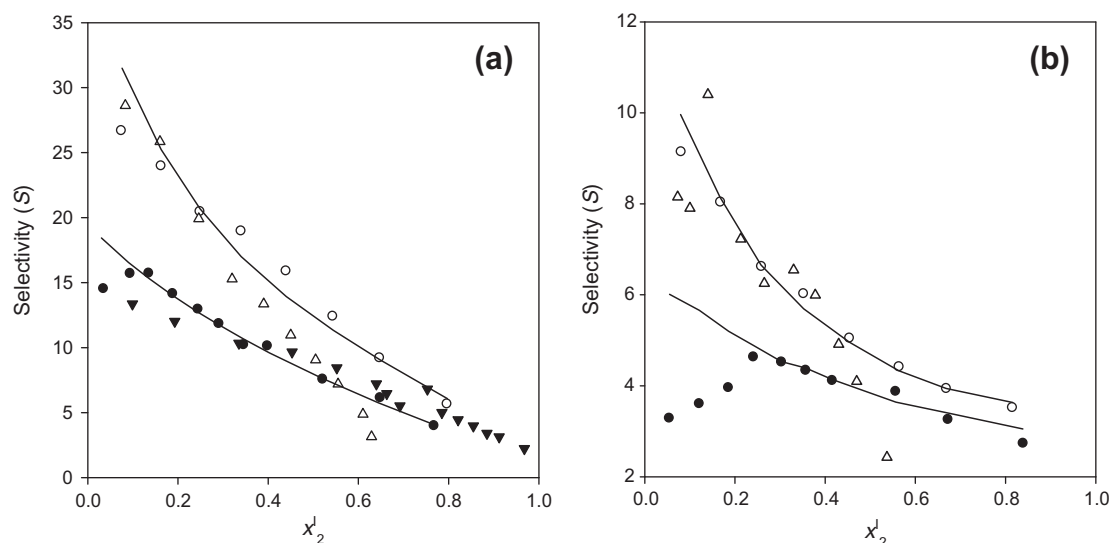


FIGURE 2. Plot of selectivity against mole fraction for the ternary systems (a) {heptane (1) + toluene (2) + solvent (3)} and (b) {cyclohexane (1) + toluene (2) + solvent (3)}: (●) [BMpyr][NTf₂]; (▼) [BMpyr][NTf₂] from reference [11]; (○) [BMpyr][TFO]; (△) sulfolane from reference [20]; solid line represents selectivity calculated with NRTL model.

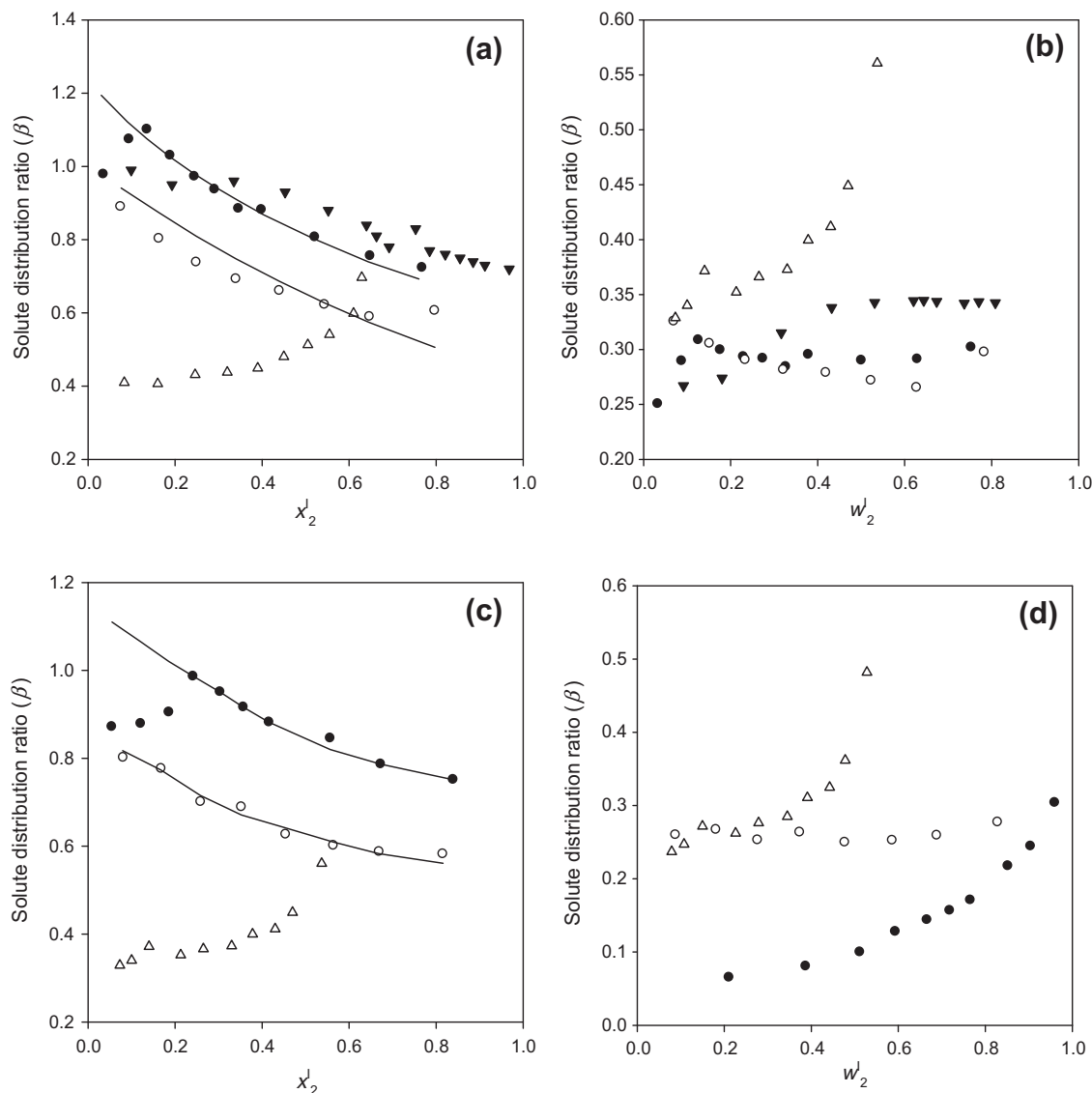


FIGURE 3. Plot of solute distribution ratio for the ternary systems {heptane (1) + toluene (2) + solvent (3)} against (a) molar fraction and (b) mass fraction; and solute distribution ratio for the ternary systems {cyclohexane (1) + toluene (2) + solvent (3)} against (c) molar fraction and (d) mass fraction: (●) [BMpyr][NTf₂]; (○) [BMpyr][TFO]; (△) sulfolane from reference [20], (▼) [BMpyr][NTf₂] from reference [11]; solid line represents solute distribution ratio calculated with NRTL model.

can be inferred from figure 3(a) and (c), the determined values of β as a function of toluene mole fraction decrease with an increase of the toluene content in the upper phase for the systems with ionic liquid, however, these values increase with an increase of toluene content for the systems containing sulfolane as was observed in several previously published studies [18–21]. The β values obtained with the IL [BMpyr][NTf₂] are higher than those obtained using [BMpyr][TFO] or sulfolane [22]. On the other hand, representing the solute distribution ratio as function of the mass fraction, figure 3(b) and (d), the determined values of β do not present remarkable variation with toluene mass fraction content in the upper phase and the β values obtained with sulfolane are higher than those obtained using the ILs. This is due to the high molar mass of the ILs, much higher than sulfolane. Besides, as well as the behavior found in the S values, NRTL model correlates satisfactorily the experimental data, with the bigger deviations found in the system {cyclohexane (1) + toluene (2) + [BMpyr][NTf₂] (3)} in the lower composition range.

Finally, the last step was to analyze the other ternary systems {heptane (1) + cyclohexane (2) + IL (3)}. As can be seen in table 3, the S and β values decrease when the content of toluene increases,

except for the system {heptane (1) + cyclohexane (2) + [BMpyr][NTf₂] (3)}, when the β values remain almost constant.

In summary, as both ILs present high values of S they can be used as solvents in extraction. On the other hand, due to the low values of β , especially in mass fraction, it would be necessary big quantities of the solvents. Still, as the ILs can be recovered and used many times, it does not represent a major problem. From the results obtained, it can be concluded that for the systems studied [BMpyr][TFO] can be considered a better solvent, since this IL presents S values higher than the ionic liquid with [NTf₂][−] anion, although the β values determined in mass fraction from experimental LLE data are similar for the systems {heptane (1) + cyclohexane (2) + IL (3)}.

3.2. Thermodynamic correlation

The correlation of the experimental (liquid + liquid) equilibrium data was carried out using the NRTL model [11]. Even though this model was created for systems involving non-electrolytes, it usually leads to good correlations for LLE data of systems containing ILs [23–26].

TABLE 4

Values of the NRTL binary interaction parameters and deviations for LLE data of the ternary systems at $T = 298.15$ K.

$i-j$	$\Delta g_{ij}/(\text{kJ} \cdot \text{mol}^{-1})$	$\Delta g_{ji}/(\text{kJ} \cdot \text{mol}^{-1})$	α_{ij}	σx	$\Delta \beta$
<i>Heptane (1) + toluene (2) + [BMpyr][NTf₂] (3)</i>					
1–2	–1.116	4.379	0.30	0.144	1.45
1–3	15.437	5.816	0.30		
2–3	11.338	–1.871	0.30		
<i>Cyclohexane (1) + toluene (2) + [BMpyr][NTf₂] (3)</i>					
1–2	–1.456	5.806	0.34	0.434	3.73
1–3	26.969	5.962	0.28		
2–3	22.799	0.526	0.40		
<i>Heptane (1) + cyclohexane (2) + [BMpyr][NTf₂] (3)</i>					
1–2	–4.539	11.251	0.15	0.132	3.21
1–3	284.232	8.571	0.15		
2–3	44.211	2.512	0.15		
<i>Heptane (1) + toluene (2) + [BMpyr][TfO] (3)</i>					
1–2	–1.5163	6.4981	0.23	0.113	0.90
1–3	12.1845	5.9025	0.26		
2–3	12.6538	–1.6243	0.24		
<i>Cyclohexane (1) + toluene (2) + [BMpyr][TfO] (3)</i>					
1–2	263.2914	–0.7024	0.15	0.153	1.21
1–3	18.5740	2.5242	0.15		
2–3	24.9738	–3.9182	0.15		
<i>Heptane (1) + cyclohexane (2) + [BMpyr][TfO] (3)</i>					
1–2	2.766	1.597	0.15	0.123	5.55
1–3	986.205	10.792	0.15		
2–3	20.803	1.056	0.15		

Table 4 summarizes the fitting parameters obtained using the NRTL model. Following a procedure based on the freedom of the non-randomness parameter α_{ij} for each ternary mixture, the best results achieved are presented in table 4. The typical values of α_{ij} are found in the range 0.2 to 0.47, but in some systems the best outcome was achieved using values outside this range. They were fixed after checking different values in the mentioned range. Also, in the systems {cyclohexane (1) + toluene (2) + [BMpyr][NTf₂]} and {heptane (1) + toluene (2) + [BMpyr][TfO]}, α_{ij} was set for each pair of components instead for the whole system. The binary interaction parameters, Δg , are estimated from experimental data. The objective function used minimizes the differences between the experimental and calculated mole fraction of the components in both phases. The values of the root mean square deviation of the composition, σx , and the mean error of the solute distribution ratio, $\Delta \beta$, were calculated using the following expressions:

$$\sigma x = 100 \left(\frac{\sum_i^M \sum_j^{N-1} \left(\left(x_{ij}^{\text{I,exp}} - x_{ij}^{\text{I,calc}} \right)^2 + \left(x_{ij}^{\text{II,exp}} - x_{ij}^{\text{II,calc}} \right)^2 \right)}{2MN} \right)^{1/2}, \quad (3)$$

$$\Delta \beta = 100 \left(\frac{1}{M} \sum_k \left(\frac{\beta_k - \beta_k^{\text{calc}}}{\beta_k} \right)^2 \right)^{1/2}, \quad (4)$$

where M is to the number of tie-lines and N refers to the number of components in the mixture.

The correlated and experimental tie-lines are drawn in figure 1 for comparison purpose. As can be concluded from this figure and the deviations shown in table 4, the NRTL model correlates satisfactorily the experimental LLE data for the studied ternary systems. Nonetheless, the model is not able to properly correlate the alkane-rich phase due to the presence of small amounts of ionic liquids in some of the systems.

4. Conclusions

Liquid–liquid experimental data were determined for the ternary systems {heptane (1) + toluene (2) + IL (3)}, {cyclohexane (1) + toluene (2) + IL (3)}, and {heptane (1) + cyclohexane (2) + IL (3)}, for the ILs [BMpyr][NTf₂] and [BMpyr][TfO] at $T = 298.15$ K. All the studied systems present high values of selectivity and values of solute distribution lower than sulfolane in mass fraction. This implies that [BMpyr][NTf₂] and [BMpyr][TfO] are suitable to replace organic solvents, because although the low values of solute distribution ratio evince that big quantities of ILs are needed, their recovery is easier and cheaper than the latter ones.

With regard to the influence of the anion of the IL on the phase equilibria, it was proved that the IL with the [TfO][–] anion presents the higher S values and since their β values are similar, [BMpyr][TfO] is the better candidate for use in extraction between the two studied.

The NRTL model was used to correlate the experimental LLE data for the six studied ternary systems, and after the inspection of the obtained values of σx and $\Delta \beta$ it can be inferred that this model correlates adequately the (liquid + liquid) equilibrium data of all the systems studied.

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